Łukasiewicz's and Klene's 3-Valued Logics | 5

5.1 Philosophical Motivations

Standard logic is built on a simple and powerful idea: every proposition is either true or false. Thus, a logical system is constructed in which every formula can be interpreted as being either true or false.

Yet many situations motivate deviating from this duality. What should we say about a statement whose truth is unknown? What about propositions concerning future contingents, incomplete information, or computations that fail to terminate?

Questions like these motivated the development of three-valued logics, systems in which a third logical value supplements the classical pair of truth and falsity.

In the early twentieth century, Jan Łukasiewicz developed one of the first systematic three-valued logics. Łukasiewicz was motivated partly by philosophical problems concerning future contingent statements, inspired by Aristotle's discussion of the issue.

In *De Interpretatione* (Book 9), Aristotle considers propositions about future contingent events, such as that there will be a sea battle tomorrow. If every proposition must already be either true or false, then it seems that either the sea battle must occur with necessity or it must fail to occur with necessity. This appears to lead to a form of fatalism: the future would already be fixed independently of human action or chance.

Aristotle and Łukasiewicz, therefore, questioned whether propositions about contingent future events should be assigned a definite truth value before the events themselves occur. Łukasiewicz regarded this problem as evidence that classical two-valued logic is not always adequate for reasoning about the future.

Moreover, he proposed introducing a third truth value, often interpreted as *possibility* or *indeterminacy*. Under this interpretation, the statement that there will be a sea battle tomorrow is neither true nor false today, but would be assigned this third value.

A different motivation emerged in the work of Stephen Cole Kleene. In connection with partial recursive functions and computations that may fail to terminate, Kleene introduced influential three-valued semantics in which the third value represents *undefinedness* (Kleene 1971, §64).

Kleene proposed two different understanding for the logical operators in this three-valued framework. We know these systems as *weak Kleene* WK and *strong Kleene* SK.

5.2 Vocabularies and Languages

The vocabularies and formal languages $\mathbf{Fo}_{\mathsf{WK,SK}}$, and $\mathbf{Fo}_{\mathsf{L3}}$ will be the same as those of the standard system.

Definition 5.1.

$$\mathbf{Vo}_{\mathsf{SK}} \overset{\text{DF}}{=} \mathbf{Vo}_{\mathsf{S0}}. \qquad \mathbf{Vo}_{\mathsf{WK}} \overset{\text{DF}}{=} \mathbf{Vo}_{\mathsf{S0}}. \qquad \mathbf{Vo}_{\mathsf{L3}} \overset{\text{DF}}{=} \mathbf{Vo}_{\mathsf{S0}}.$$

$$\mathbf{Fo}_{\mathsf{SK}} \overset{\text{DF}}{=} \mathbf{Fo}_{\mathsf{S0}}. \qquad \mathbf{Fo}_{\mathsf{WK}} \overset{\text{DF}}{=} \mathbf{Fo}_{\mathsf{S0}}. \qquad \mathbf{Fo}_{\mathsf{L3}} \overset{\text{DF}}{=} \mathbf{Fo}_{\mathsf{S0}}.$$

An alternative way would be to define a language including additional logical operators which behave differently to the classical ones. However, we will here take the usual approach of giving an alternative account of the usual logical operators.

5.3 Inference and Consequence Relations

WK, SK, and L3 fall among modern systems, so their inference relations are just cases of definition 3.31.

As for the contents of their consequence relations, we want the law of excluded third to be not valid in general. In other words, we do not want that all inferences of the following form be members of their consequence relations:

1. $/ \; (\mathtt{A} \vee \neg\mathtt{A})$

5.4 Interpretation Systems

In both WK and SK, the non-logical symbols are propositional variables. A three-valued propositional interpretation, or 3pi for short, assigns to each propositional variable a logical truth value among $\mathbf{1}$ (truth), $\frac{1}{2}$ (the third value), $\mathbf{0}$ (falsity).

Definition 5.2 (Three-valued propositional interpretation)**.**

$$\langle \mathbf{D}, \iota \rangle \text{ is a } \mathbf{3pi} \text{ on } \mathbf{Fo}_{\mathsf{S}}$$
$$\overset{\text{DF}}{=} \mathbf{D} = \{\mathbf{0}, \tfrac{1}{2}, \mathbf{1}\} \mathbin{\wedge\!\!\wedge} \iota : \mathbf{Pro}_{\mathsf{S}} \longrightarrow \{\mathbf{0}, \tfrac{1}{2}, \mathbf{1}\}.$$

In other words, $\langle \mathbf{D}, \iota \rangle$ is a 3pi on $\mathbf{Fo}_{\mathsf{S}}$ iff its domain is the set of logical values $\{\mathbf{0}, \frac{1}{2}, \mathbf{1}\}$ and its interpretation map ι is a function from the set of propositional variables to $\{\mathbf{0}, \frac{1}{2}, \mathbf{1}\}$ – that is, each propositional variable is assigned either just $\mathbf{0}$ or just $\frac{1}{2}$ or just $\mathbf{1}$.

A 3pi $\mathfrak{I}$ is extended to all formulae of $\mathbf{Fo}_{\mathsf{S}}$ by means of a valuation.

We have two kinds of extensions of a 3pi, related to the two systems we are studying: weak 3-valued extensions (w3e) for WK, strong 3-valued extensions (s3e) for SK, and Łukasiewicz 3-valued extensions (Ł3e) for L3.

In the following, note that the notion of semantic consequence we need for all these logics is standard, thus, it satisfies definition 3.39. Moreover, the designated value of all these logics will be $\mathbf{1}$, while the others will be anti-designated.

Weak Kleene WK**.** A weak 3-valued extensions (w3e) on a 3pi is defined as follows.

Definition 5.3 (Weak 3-valued propositional valuation)**.** A valuation $\hat{\mathfrak{I}}$ is a **w3e** of the 3pi $\mathfrak{I}$ on $\mathbf{Fo}_{\mathsf{S}}$ $\overset{\text{DF}}{=}$ $\hat{\iota}(\mathtt{A}) = \iota(\mathtt{A}) \mathbin{\wedge\!\!\wedge}$

$\mathtt{F}$	$\neg\mathtt{F}$
$\mathbf{0}$	$\mathbf{1}$
$\frac{1}{2}$	$\frac{1}{2}$
$\mathbf{1}$	$\mathbf{0}$

$\wedge$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$
$\mathbf{0}$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{0}$
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$\mathbf{1}$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$

$\vee$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$
$\mathbf{0}$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$\mathbf{1}$	$\mathbf{1}$	$\frac{1}{2}$	$\mathbf{1}$

$\rightarrow$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$
$\mathbf{0}$	$\mathbf{1}$	$\frac{1}{2}$	$\mathbf{1}$
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$\mathbf{1}$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$

$\leftrightarrow$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$
$\mathbf{0}$	$\mathbf{1}$	$\frac{1}{2}$	$\mathbf{0}$
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$\mathbf{1}$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$

$: \mathtt{A} \in \mathbf{Pro}_{\mathsf{S}} \mathbin{\wedge\!\!\wedge} \mathtt{F} \in \mathbf{Fo}_{\mathsf{S}}.$

Note that weak extensions are obtained by filling the additional cells in the resulting truth tables with '$\frac{1}{2}$', which stands for the third value.

$\mathbf{Sem}_{\mathsf{WK}}$ is defined based on definition 5.3 as follows:

Definition 5.4 ($\mathbf{Sem}_{\mathsf{WK}}$)**.**

$\mathbf{Sem}_{\mathsf{WK}}$ is a semantics on $\mathbf{Fo}_{\mathsf{WK}}$

$$\begin{aligned}
&\overset{\text{DF}}{=} \mathbf{Int}_{\mathsf{WK}} = \{\mathfrak{I} : \mathfrak{I} \text{ is a 3pi on } \mathbf{Fo}_{\mathsf{WK}}\} && \wedge\!\!\wedge\\
&\mathbf{LV}_{\mathsf{WK}} = \{0, \tfrac{1}{2}, 1\} && \wedge\!\!\wedge\\
&\mathbf{Des}^{+}_{\mathsf{WK}} = \{1\} && \wedge\!\!\wedge\\
&\mathbf{Des}^{-}_{\mathsf{WK}} = \{\tfrac{1}{2}, 0\} && \wedge\!\!\wedge\\
&\mathbf{Val}_{\mathsf{WK}} = \{\hat{\mathfrak{I}} : \exists\mathfrak{I} \in \mathbf{Int}_{\mathsf{WK}}\,(\hat{\mathfrak{I}} \text{ is a w3e of } \mathfrak{I})\} && \wedge\!\!\wedge\\
&\vDash_{\mathsf{WK}} \subseteq \mathbf{In}_{\mathsf{WK}} \wedge\!\!\wedge \vDash_{\mathsf{WK}} \text{ is classical.}
\end{aligned}$$

Strong Kleene SK**.** A strong 3-valued extensions (s3e) on a 3pi is defined as follows.

Definition 5.5 (Strong 3-valued propositional valuation)**.** A valuation $\hat{\mathfrak{I}}$ is a **s3e** of the 3pi $\mathfrak{I}$ on $\mathbf{Fo}_{\mathsf{S}} \overset{\text{DF}}{=} \hat{\iota}(\mathtt{A}) = \iota(\mathtt{A}) \wedge\!\!\wedge$

$\mathtt{F}$	$\neg\mathtt{F}$
0	1
$\frac{1}{2}$	$\frac{1}{2}$
1	0

$\wedge$	0	$\frac{1}{2}$	1
0	0	0	0
$\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{2}$
1	0	$\frac{1}{2}$	1

$\vee$	0	$\frac{1}{2}$	1
0	0	$\frac{1}{2}$	1
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1
1	1	1	1

$\rightarrow$	0	$\frac{1}{2}$	1
0	1	1	1
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1
1	0	$\frac{1}{2}$	1

$\leftrightarrow$	0	$\frac{1}{2}$	1
0	1	$\frac{1}{2}$	0
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1	0	$\frac{1}{2}$	1

: $\mathtt{A} \in \mathbf{Pro}_{\mathsf{S}} \wedge\!\!\wedge \mathtt{F} \in \mathbf{Fo}_{\mathsf{S}}$.

Strong extensions differ from weak extensions with regard to the value assigned to formulae with '$\wedge$', '$\vee$', '$\rightarrow$'. The truth-tables of such formulae are obtained by filling the additional cells in the resulting truth tables according to the criteria:

- There is an ordering $0 < \frac{1}{2} < 1$.
- A disjunction preserves the higher logical value of its disjuncts.
- A conjunction preserves the lower logical value of its conjuncts.
- A conditional is assigned 1 when the antecedent is assigned 0 or its consequent is assigned 0; it is assigned 0 when the antecedent is assigned 1 and the consequent is assigned 0; otherwise, it is assigned $\frac{1}{2}$.

$\mathbf{Sem}_{\mathsf{SK}}$ is defined based on definition 5.5 as follows:

Definition 5.6 ($\mathbf{Sem}_{\mathsf{SK}}$)**.**

$\mathbf{Sem}_{\mathsf{SK}}$ is a semantics on $\mathbf{Fo}_{\mathsf{SK}}$

$\overset{\text{DF}}{=} \mathbf{Int}_{\mathsf{SK}} = \{\mathfrak{I} : \mathfrak{I} \text{ is a 3pi on } \mathbf{Fo}_{\mathsf{SK}}\}$ ⩕

$\mathbf{LV}_{\mathsf{SK}} = \{\mathbf{0}, \frac{1}{2}, \mathbf{1}\}$ ⩕

$\mathbf{Des}^{+}_{\mathsf{SK}} = \{\mathbf{1}\}$ ⩕

$\mathbf{Des}^{-}_{\mathsf{SK}} = \{\frac{1}{2}, \mathbf{0}\}$ ⩕

$\mathbf{Val}_{\mathsf{SK}} = \{\hat{\mathfrak{I}} : \exists \mathfrak{I} \in \mathbf{Int}_{\mathsf{SK}} (\hat{\mathfrak{I}} \text{ is a s3e of } \mathfrak{I})\}$ ⩕

$\vDash_{\mathsf{SK}} \subseteq \mathbf{In}_{\mathsf{SK}}$ ⩕ $\vDash_{\mathsf{SK}}$ is classical.

Łukasiewicz 3-valued logic $\mathsf{Ł3}$**.** A Łukasiewicz 3-valued extensions (Ł3e) on a 3pi is defined as follows.

Definition 5.7 (Łukasiewicz 3-valued propositional valuation)**.** A valuation $\hat{\mathfrak{I}}$ is a **Ł3e** of the 3pi $\mathfrak{I}$ on $\mathbf{Fo}_{\mathsf{S}} \overset{\text{DF}}{=} \hat{\iota}(\mathtt{A}) = \iota(\mathtt{A})$ ⩕

F	¬F
$\mathbf{0}$	$\mathbf{1}$
$\frac{1}{2}$	$\frac{1}{2}$
$\mathbf{1}$	$\mathbf{0}$

$\wedge$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$
$\mathbf{0}$	$\mathbf{0}$	$\mathbf{0}$	$\mathbf{0}$
$\frac{1}{2}$	$\mathbf{0}$	$\frac{1}{2}$	$\frac{1}{2}$
$\mathbf{1}$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$

$\vee$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$
$\mathbf{0}$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\mathbf{1}$
$\mathbf{1}$	$\mathbf{1}$	$\mathbf{1}$	$\mathbf{1}$

$\rightarrow$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$
$\mathbf{0}$	$\mathbf{1}$	$\mathbf{1}$	$\mathbf{1}$
$\frac{1}{2}$	$\frac{1}{2}$	$\mathbf{1}$	$\mathbf{1}$
$\mathbf{1}$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$

$\leftrightarrow$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$
$\mathbf{0}$	$\mathbf{1}$	$\frac{1}{2}$	$\mathbf{0}$
$\frac{1}{2}$	$\frac{1}{2}$	$\mathbf{1}$	$\frac{1}{2}$
$\mathbf{1}$	$\mathbf{0}$	$\frac{1}{2}$	$\mathbf{1}$

: $\mathtt{A} \in \mathbf{Pro}_{\mathsf{S}}$ ⩕ $\mathtt{F} \in \mathbf{Fo}_{\mathsf{S}}$.

Łukasiewicz extensions differ from weak extensions with regard to the value assigned to formulae with '→', '↔'. In these truth tables, we replace '$\frac{1}{2}$' in the cell $(\frac{1}{2}, \frac{1}{2})$ with '$\mathbf{1}$'.*

The logic behind this decision is, again, to assume an ordering $\mathbf{0} < \frac{1}{2} < \mathbf{1}$. Hence, we assign the conditional $\mathbf{1}$ in an interpretation iff the antecedent and the consequent are assigned values $\mathbf{v}$ and $\mathbf{w}$, respectively, such that $\mathbf{v} \leq \mathbf{w}$.

*: Another version of Łukasiewicz's system includes additional operators to represent modal notions such as possibility, necessity, and contingency, with semantics which are very alien to the treatment of these notions in modern modal logics.

$\mathbf{Sem}_{\text{Ł3}}$ is defined based on definition 5.7 as follows:

Definition 5.8 ($\mathbf{Sem}_{\text{Ł3}}$).

$\mathbf{Sem}_{\text{Ł3}}$ is a semantics on $\mathbf{Fo}_{\text{Ł3}}$

$$\begin{aligned}
\overset{\text{DF}}{=}\ & \mathbf{Int}_{\text{Ł3}} = \{\mathfrak{I} : \mathfrak{I} \text{ is a 3pi on } \mathbf{Fo}_{\text{Ł3}}\} && \wedge\!\!\!\wedge \\
& \mathbf{LV}_{\text{Ł3}} = \{\mathbf{0}, \tfrac{1}{2}, \mathbf{1}\} && \wedge\!\!\!\wedge \\
& \mathbf{Des}^{+}_{\text{Ł3}} = \{\mathbf{1}\} && \wedge\!\!\!\wedge \\
& \mathbf{Des}^{-}_{\text{Ł3}} = \{\tfrac{1}{2}, \mathbf{0}\} && \wedge\!\!\!\wedge \\
& \mathbf{Val}_{\text{Ł3}} = \{\hat{\mathfrak{I}} : \exists \mathfrak{I} \in \mathbf{Int}_{\text{Ł3}}\, (\hat{\mathfrak{I}} \text{ is a Ł3e of } \mathfrak{I})\} && \wedge\!\!\!\wedge \\
& \vDash_{\text{Ł3}} \subseteq \mathbf{In}_{\text{Ł3}} \wedge\!\!\!\wedge \vDash_{\text{Ł3}} \text{ is classical.}
\end{aligned}$$

Results As desired, '$(p \vee \neg p)$' is not a member of any of these semantic consequence relations.

Theorem 5.9. $\nvDash_{\text{WK}} (p \vee \neg p) \wedge\!\!\!\wedge \nvDash_{\text{SK}} (p \vee \neg p) \wedge\!\!\!\wedge \nvDash_{\text{Ł3}} (p \vee \neg p)$.

Proof. Consider the following truth table:

p	$(p$	$\vee$	$\neg p)$
$\mathbf{0}$		$\mathbf{1}$	$\mathbf{1}$
$\frac{1}{2}$		$\frac{1}{2}$	$\frac{1}{2}$
$\mathbf{1}$		$\mathbf{1}$	$\mathbf{0}$

Note that the second row of this table corresponds to a 3pi $\left\langle\{\mathbf{0}, \frac{1}{2}, \mathbf{1}\}, \iota\right\rangle$ with a 3pe $\hat{\mathfrak{I}}$ such that $\hat{\iota}(\text{'}(p \vee \neg p)\text{'}) = \frac{1}{2}$. But since $\frac{1}{2}$ is in $\mathbf{Des}^{-}_{\text{WK}}$, $\mathbf{Des}^{-}_{\text{SK}}$, and $\mathbf{Des}^{-}_{\text{Ł3}}$, we have that $\nvDash_{\text{WK}} (p \vee \neg p)$, $\nvDash_{\text{SK}} (p \vee \neg p)$, and $\nvDash_{\text{Ł3}} (p \vee \neg p)$. □

Task 5.1. Show whether the following inferences are members of $\models_{\mathsf{WK}}$, $\models_{\mathsf{SK}}$, and $\models_{\mathsf{L3}}$.

1. $q \; / \; (p \rightarrow q)$
2. $\neg p \; / \; (p \rightarrow q)$
3. $((p \wedge q) \rightarrow r) \; / \; ((p \rightarrow r) \vee (q \rightarrow r))$
4. $((p \rightarrow q) \wedge (r \rightarrow s)) \; / \; ((p \rightarrow s) \vee (r \rightarrow q))$
5. $\neg(p \rightarrow q) \; / \; p$
6. $(p \rightarrow r) \; / \; ((p \wedge q) \rightarrow r)$
7. $(p \rightarrow q), (q \rightarrow r) \; / \; (p \rightarrow r)$
8. $(p \rightarrow q) \; / \; (\neg p \rightarrow \neg q)$
9. $p \; / \; (q \vee \neg q)$
10. $/ \; (p \rightarrow (q \vee \neg q))$
11. $p, \neg p \; / \; q$
12. $(p \wedge \neg p) \; / \; q$
13. $/ \; ((p \wedge \neg p) \rightarrow q)$
14. $(p \rightarrow q), p \; / \; q$
15. $(p \vee q), \neg p \; / \; q$
16. $/ \; (p \vee \neg p)$
17. $/ \; \neg\neg(p \vee \neg p)$
18. $\neg\neg p \; / \; p$
19. $p \; / \; \neg\neg p$
20. $\neg\neg\neg p \; / \; \neg p$

5.5 Derivation Systems

It is perfectly possible to provide derivation systems for three-valued logics. We will not do it here, as their semantics provide the essence of these logics.